FI SHARIF (X) 3297

EXXON

SUPPORTING STRUCTURES AND FOUNDATIONS FOR HEAVY MACHINERY

A10

SCOPE

I 1.1 This practice covers the design and testing of supporting structures and foundations for heavy machinery.

I 1.2 An asterisk (*) indicates that additional information is required. If a job is contracted, this additional information is provided in the Job Specification.

I 1.3 ACI Standard 207, Mass Concrete, shall be used with this practice.

DEFINITIONS

I 2.1 **Heavy machinery** is any equipment having reciprocating or rotary masses as the major moving parts (such as compressors, pumps, engines, and turbines), and having a total weight greater than 5,000 pounds (2270 kg).

DESIGN

ALL HEAVY MACHINERY

I 3.1 **Dynamic modulus of elasticity** of concrete (E') for use in dynamic analyses shall be:

$E = 79{,}000\sqrt{f'_c}$ where f'_c = 28 day standard cylinder compressive strength, psi

$(E' = 6{,}550\sqrt{f'_c}$, f'_c in MPa)

R 3.2 **Soil bearing pressure** shall not exceed 50 percent of the net allowable values for static loads. For piled foundations, no reduction in allowable pile capacity is required.

R 3.3 The effects of **shrinkage and thermal expansion** shall be taken into account. To prevent cracking, minimum concrete reinforcing shall be 84 lb/cu yd (50 kg/m^3), except for the foundation slab, which shall be at least 50 lb/cu yd (30 kg/m^3). Reinforcing steel shall be placed on all faces of the foundation base slab and/or block. An intermediate layer of two-way reinforcing is required when the thickness exceeds 5 ft. (1.5 m). Spacing between horizontal layers shall not exceed 3.25 ft (1 m). The interior of blocks with least horizontal dimension more than 5 ft (1.5 m) shall be reinforced with vertical bars spaced no more than 3.25 ft (1 m) between bars.

R 3.4 To prevent **fatigue failures,** all sections shall be proportioned to resist the sum of: static dead and live loads, plus **three (3) times dynamic loads.** For reciprocating machine foundations, dynamic loads shall include individual piston forces.

O 3.5 **Foundation design** shall consist of clean simple lines. Beams, columns and slabs shall be of uniform rectangular shapes. Pockets where vapors could accumulate are not permitted. Provision shall be made to drain oil and rainwater from under the equipment. Sufficient space shall be provided for installation, maintenance, and operation; piping and anchor bolts.

O 3.6 All parts of machine supports shall be isolated from adjacent foundations and buildings to prevent vibration transmission.

a. **Auxiliary structures** such as access platforms and cranes shall be supported independently of the machine supports and foundations.

b. The machine foundation shall be spaced a minimum of 1/2 inch (12 mm) from adjacent foundations and floor slabs. The space between the two shall be filled with a flexible joint filler and sealer.

R 3.7 The thickness of the foundation slab, in feet (m), shall not be less than:

$$2 + \frac{L}{30} \qquad \left(0.6 + \frac{L}{30}\right)$$

where: For one machinery train:

L = longest dimension of the foundation slab, ft (m)

For two or more machinery trains supported off a common foundation:

L = greater of: **the width** of the common slab, or,

the length of the longest slab segment assigned to any one train, ft (m)

O 3.8 The height of support above grade shall be the minimum required to accommodate suction and discharge piping configuration and shall be at least 8 inches (200 mm) above finished grade.

I 3.9 Soil properties shall be based on the results of a soils investigation.

R 3.10 The foundation system shall be designed considering a range of soil shear modulus values from 0.5 to 1.5 times the nominal value. If the nominal soil shear modulus is determined by a direct soil testing method acceptable to the owner's engineer such as a cross-hole field measurement or resonant column test, the range may be reduced to 0.75 to 1.25 times the nominal value.

DESIGN (Cont)

RECIPROCATING MACHINERY

R 4.1 To prevent thermally-induced cracking during hydration, concrete construction shall be per **ACI 207 Mass Concrete** provisions.

R 4.2 **Reciprocating machinery** and associated pulsation suppression devices shall be supported directly on a **rigid foundation.** The support foundation for the pulsation suppression devices shall be integral with the compressor foundation. Mounting plates shall be anchored into the foundation. Foundation design shall minimize re-entrant type corners and other details causing stress concentrations.

R 4.3 **Rigid machinery foundations** shall be designed as follows:

a. The horizontal eccentricity, in any direction, between the centroid of mass of the machine-foundation system and the centroid of the base contact area, shall not exceed 5% of the respective base dimension.

b. The center of gravity of the machine-foundation system shall be as close as possible to the lines of action of the unbalanced forces.

c. Groups of reciprocating machinery shall be tied together with a common foundation slab when the combined foundation results in reduced amplitudes.

R 4.4 **Static design** shall take into account the following:

a. Weight of all machines on the foundation.

b. Unbalanced forces and couples, as specified by the machine manufacturer.

c. Individual piston forces

R 4.5 **Dynamic design** shall be as follows:

a. For balanced, opposed reciprocating compressors, a **lateral force** equal to 25 percent of the piston rod load shall be applied to the foundation between the crankcase and cross-head housing support.

b. **Primary forces, couples and moments** shall be applied at machine speed, over the full range of specified operating speeds, for calculation of primary amplitudes.

c. **Secondary forces, couples and moments** shall be applied at twice machine speed, over the full range of specified operating speeds, for calculation of secondary amplitudes.

d. **Total amplitudes** shall be calculated by combining, in phase, primary and secondary amplitudes per subpar. b. and c. above. No total peak-to-peak amplitude on the foundation shall exceed 0.002 in. (0.05/mm).

e. **Damping** shall be considered in amplitude calculations if ratios of natural frequencies to disturbing frequencies in the range of 0.7 to 1.3 cannot be avoided in the modes being excited: Geometric Damping, when used, shall be calculated consistent with the principals of the "Elastic Half-Space Model" Theory. The geometric damping ratio shall not exceed 2/3 of the theoretical value nor 0.7 for all modes.

ROTARY MACHINERY

I 5.1 **Rotary machinery** may be **supported** either directly on a **rigid (block type) foundation**, or on an **elevated structure.**

R 5.2 **Static design** for either type foundation shall take into account the following loads:

a. The dead weight of the machines and their baseplates.

b. Vertical forces representing 50% of the weight of each machine, including its baseplate, applied normal to its shaft at a point midway between the end bearings.

c. Lateral forces representing 25% of the weight of each machine, including its baseplate, applied normal to its shaft at a point midway between the end bearings.

d. Longitudinal forces representing 25% of the weight of each machine, including its baseplate, applied along the shaft axis.

e. The total vertical, lateral and longitudinal forces per subpar. b, c, and d above, shall not be considered to act concurrently.

R 5.3 **Dynamic design** for either type foundation shall be as follows:

a. Amplitudes shall be determined using dynamic forces from each rotor, calculated as follows:

$$\text{Dynamic force} = \frac{(\text{rotor weight}) \times (\text{rotor speed, rpm})}{6000}$$

DESIGN (Cont)

b. When there is more than one rotor, amplitudes shall be computed with the rotor forces assumed in-phase and 180° out-of-phase to obtain, respectively, the maximum translational and torsional amplitudes.

c. For the design of the foundation, the machinery shall be considered as a rigid body attached to the foundation system. Machinery flexibility shall not be included in the calculation and evaluation of the total peak-to-peak amplitudes at the center line of the shaft of the machine, unless otherwise specified by the Owner's Engineer.

d. The total calculated peak-to-peak amplitude on any point of the structure, and/or foundation system above grade (including points along the center line of the shaft of the machine) shall not exceed the following values:

SPEED OF ROTOR, rpm	MAXIMUM ALLOWABLE PEAK-TO-PEAK AMPLITUDE, in. (mm)	
0 – 999	0.0014	(0.035)
1000 – 1169	0.0013	(0.033)
1170 – 1269	0.0012	(0.030)
1270 – 1389	0.0011	(0.028)
1390 – 1525	0.0010	(0.025)
1526 – 1699	0.0009	(0.023)
1700 – 1899	0.0008	(0.020)
1900 – 2179	0.0007	(0.018)
2180 – 2499	0.0006	(0.015)
2500 and above-	0.0005	(0.013)

1 MIL = (1/1000) × INCH
= 0.001 INCH
= 0.025 MM

R 5.4 **Rigid foundations** for rotary machinery shall be designed per par. 4.3 a. and b. and par. 4.5 e.

R 5.5 **Elevated structures** for rotary machinery shall be designed as follows:

a. **Machine loads** shall be directly over vertical supports.

b. **Beams and slabs** shall have a minimum span.

c. The **upper table** and the **foundation slab** shall be rigid in the horizontal plane.

d. The **weight** of the upper table plus 1/2 of the weight of the columns shall be greater than the weight of the machine(s).

e. The foundation slab shall not weigh less than the combined supported weight of the upper table, columns or walls, and the machines (including baseplates).

f. The effects of short circuit couples, faulty synchronizing conditions, oil whirl frequency, rotor critical speeds, and background vibration shall be considered.

g. Transverse frames or walls shall have the same vertical natural frequencies, within ±5%.

h. Torsional and transverse horizontal natural frequencies shall be determined considering the whole structure. Individual transverse bents or walls shall have the same transverse horizontal natural frequencies, within ±5%.

i. Multi-degrees of freedom shall be considered for frames if a single degree of freedom system will not lead to an acceptable mathematical representation of the structure. A multi-degree of freedom analysis shall be used for foundations with walls.

j. Loaded beam and slab natural frequencies in both the horizontal and vertical directions, where possible, shall be above any machine speed. If beams or slabs must be designed to have natural frequencies below machine speed, allowance must be made for the stiffening effect of the baseplates and the machine.

k. For analyses based on one or two degree of freedom models, all natural frequencies shall be out of the range of 0.7 to 1.3 times each operating speeds of any machine supported thereon.

l. For multi-degrees of freedom analyses, frequencies associated with modes with effective mass greater than 5% of the total mass of the model shall be out of the range of 0.7 to 1.3 times the operating speeds.

R 5.6 **Pile Foundations for Machinery shall be designed as follows:**

a. The pile cap shall not weigh less than 2.0 times and 3.5 times the weight of the rotary and reciprocating machines, respectively.

b. The piles shall be arranged so that the centroid of the combined machine-foundation system coincides within 5% of the respective overall dimensions of the pile group.

DESIGN (Cont)

c. The geometric damping ratio for pile groups shall not exceed:
0.10 for Rocking Modes
0.15 for Horizontal, Vertical, and Torsional Modes

TESTING

6.1 A test program to determine the dynamic response of elevated supporting structures is required.

6.2 The test program shall be sufficient to determine if natural frequencies exist near machine operating speeds and to predict the approximate operating amplitudes.

*6.3 Proposed testing procedures shall be given to the Owner's Engineer for approval, at least 60 days prior to testing.

*6.4 The test program may be reduced, or waived, pending review by the Owner's Engineer of the dynamic analysis.

6.5 Testing is to commence immediately after mounting the machinery, and as long before startup as possible, in order to provide the time required to initiate corrective measures if vibration problems are discovered.

Revision Memo

9/68	Original Issue
6/69	Revision 1
1/73	Revision 2
12/90	Revision 3
12/93	Rev. 4
6/94	**Revision 0 – Original Issue of International Practice**

Added new paragraph 5.3c specifying that amplitude limits only apply to foundations and not machines. Paragraph 5.3d modified to delete that amplitude limits apply to machines.

Paragraphs renumbered

Old Par. No.	New Par. No.
5.3c	5.3d